Heart Rate Zones – Evidence Base for the App

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Bottom line

Use heart rate reserve zones by default, let the user override with measured values, and treat every HR number as an estimate.

1. **Max HR:** default to Tanaka (208 − 0.7 × age). Any formula can be off by about 20 bpm for one person, so a measured max always wins.
2. **Zones:** heart rate reserve (Karvonen) beats % of max HR. Threshold-based zones beat both, but need testing the app can't do.
3. **Effort fallback:** the talk test tracks VT1 and VT2 closely and needs no device. Use it for anyone without HR or on HR-lowering medication.
4. **Accuracy:** wrist HR during steady running is within about ±10 bpm of ECG. Smooth readings and ignore the first part of each interval.
5. **C25K focus:** finishing the intervals matters more than zone precision. Show zones as guidance, alert only on sustained high HR.

Evidence retrieved from PubMed, mostly 2021–2026.

Estimating max HR

No age-based equation predicts an individual's max HR well: average error is small, but any one person can be off by roughly 18 to 24 bpm. A measured max beats every formula.

Equation	Formula	What recent studies found
Fox	220 − age	Wide individual error, but the only formula without proportional bias in two adult samples (Shookster 2020; Martin 2025)
Tanaka	208 − 0.7 × age	Among the lowest average error in adults (Martin 2025)
Gellish	207 − 0.7 × age	Similar to Tanaka (Martin 2025)
Nes	211 − 0.64 × age	Differed significantly from measured max in a general population (Shookster 2020)

Key numbers

- 230 adults, 7 equations: mean bias −3 to +6 bpm, limits of agreement about ±18 to 24 bpm; none accurate at the individual level (Martin 2025).
- 4,043 runners: prediction error (RMSE) 9.1 to 10.5 bpm; formulas more often underestimated max HR; lab testing preferred (Kasiak 2023).
- 99 treadmill tests: all 9 equations showed poor agreement with measured max; formulas underestimated high-max people and overestimated low-max people, except Fox (Shookster 2020).

For the app: default to Tanaka, but always let the user overwrite it with a measured or observed max. If Garmin Connect shows a detected max HR from a hard effort, use that.

Calculating zones

Heart rate reserve (Karvonen) is the better of the two simple methods, but threshold-based zones beat both in randomised trials. Without lab testing, the app should use HRR and let the user correct it.

Method	How it works	Evidence
% of max HR	Zone = fixed % of HRmax	Weakest. In 163 heart failure patients, VT1 sat at 82% of peak HR, and %HRmax put only 25% in the right intensity domain (Sá Couto 2025)
% of HR reserve (Karvonen)	Target = HRrest + % × (HRmax − HRrest)	Better: 46% correctly classified in the same study. ACSM recommends 30–40% to 89% HRR (cited in Gaskill 2023)
Ventilatory or lactate threshold	Zones anchored to VT1/VT2 or LT1/LT2 from a test	Best. Three RCTs in inactive adults found larger fitness gains than HRR-based prescription (Martini 2022; Kirton 2022; Weatherwax 2024)

Karvonen formula

```
Target\ HR = HR_{rest} + \%intensity \times (HR_{max} - HR_{rest})
```

Why fixed percentages miss: threshold position varies a lot between people. In 863 exercise tests, VT ranged from about 43% to 75% of VO2 reserve depending on fitness, with wide spread at every fitness level (Gaskill 2023).

Practical anchor without a lab: mean perceived exertion at VT was 12.5 on the 6–20 Borg scale, regardless of fitness (Gaskill 2023). That is roughly "somewhat hard", the point where full sentences get difficult.

Response rates in trials (threshold vs HRR)

- VT2 improved in 79% vs 47% of participants over 12 weeks (Martini 2022)
- VO2max responders: 76% vs 21% over 13 weeks (Weatherwax 2024)
- Fitness-fatness index rose 15% vs 10% over 12 weeks on a treadmill (Kirton 2022)

What each zone trains

Physiology recognises three intensity domains split by two thresholds; the popular 5-zone model is a convenience layer on top. Research on endurance training reports zones in the 3-domain form (Sperlich 2023; Campos 2021).

5-zone label	Domain	Threshold boundary	Main training effect	Feel / talk test
Z1 Recovery	Moderate	Below VT1 / LT1	Recovery, blood flow, habit building	Full conversation
Z2 Aerobic base	Moderate	Up to VT1 / LT1	Mitochondrial and capillary growth, fat use, aerobic base	Can talk comfortably; last "yes" on the talk test ≈ VT1
Z3 Tempo	Heavy	VT1 to VT2	Raises lactate threshold; sustained hard effort	Short sentences only
Z4 Threshold	Heavy to severe	Near VT2 / LT2	Raises speed at threshold; high fatigue cost	Few words; "no" on talk test ≈ VT2
Z5 Max	Severe	Above VT2	VO2max, top-end capacity	Can't talk

The training-effect column is standard exercise physiology, not drawn from the studies cited here. Garmin's default zones (50–60, 60–70, 70–80, 80–90, 90–100% of max HR) are a manufacturer convention, not a research standard.

How much time in each zone

- In 175 elite training distributions, 91% had more than 60% of training at low intensity (Sperlich 2023).

- Runner review: about 70% or more at low intensity and 30% or less at threshold and above gave the best performance gains (Campos 2021).
- In lower-level athletes, no distribution model beat the others; polarised and pyramidal mattered mainly for elites (Rivera-Köfler 2024).

Talk test as a zone check: the last stage where speaking is comfortable matched VT1, and the stage where speaking was not comfortable matched VT2, on HR and speed (Festino 2026; Vieira 2022 systematic review; Kwon 2025).

Limits the app must handle

Wrist HR is good enough for steady treadmill running, but expect ±10 bpm noise, a lag at each interval change, and upward drift over a session.

Limit	Evidence	App response
Wrist optical HR error	62 men, 4 brands incl. Garmin: HR during endurance exercise matched ECG, limits of agreement about ±10 bpm (Lee 2026). A wrist Garmin was less accurate than forearm or upper-arm sensors during warm-up and intense full-body moves (Moghaddam 2025)	Smooth HR (rolling 5 s average); don't flash zone changes on single readings
Individual bias	Another smartwatch on a treadmill test: mean bias −2.7 bpm, limits −17 to +12 bpm (Inoue 2026)	Allow a manual offset setting if a chest strap is ever compared
Tattoos under the sensor	Tattooed skin: 5–23% error depending on activity (Navalta 2025)	Note in settings help
Cardiac drift	HR rose 14 bpm over 60 min at steady effort (Laginestra 2023); 18 bpm over about 90 min in heat (Mulholland 2023)	Warn that later minutes read higher at the same effort; keep the treadmill room cool
Beta-blockers	Blocked HR drift completely in healthy men (Laginestra 2023)	Settings flag: "On HR-lowering medication" switches the app to effort and talk-test mode

Limit	Evidence	App response
Max HR uncertainty	±18–24 bpm individual error (Martin 2025)	Let the user set a measured max

Interval lag (design choice, not from cited studies): HR takes time to rise at the start of each run interval. The app should judge zones only after the first 30–60 seconds of an interval.

App logic

Three zone modes, in order of preference. The user picks one per profile.

1. **Custom zones (best available):** user types in zone boundaries, e.g. copied from Garmin Connect or from a lab or field test.
2. **Heart rate reserve (default):** Karvonen with Tanaka max HR, unless the user enters a measured max.
3. **Effort mode:** no HR. Talk test prompt plus 1–10 effort. Used for the wife's profile and anyone on HR-lowering medication.

HRR zone boundaries (default)

Zone	% HRR	Basis
Z1	below 40%	ACSM very light to light (category values from ACSM guidelines; not opened for this doc)
Z2	40–59%	ACSM moderate
Z3	60–74%	ACSM vigorous, lower part
Z4	75–89%	ACSM vigorous, upper part; adjusted VT2 was 77% HRR in a cardiometabolic sample (Milani 2024)
Z5	90% and above	ACSM near-maximal

The Z3/Z4 split at 75% is a convenience boundary, not a validated threshold.

Code

```
const tanakaMax = age => Math.round(208 - 0.7 * age);

function hrrZones({ age, restHR, maxHR }) {
  const max = maxHR ?? tanakaMax(age);   // measured max wins
  const reserve = max - restHR;
  const at = pct => Math.round(restHR + pct * reserve);
  return [
    { zone: 1, low: restHR,  high: at(0.40) - 1 },
    { zone: 2, low: at(0.40), high: at(0.60) - 1 },
    { zone: 3, low: at(0.60), high: at(0.75) - 1 },
    { zone: 4, low: at(0.75), high: at(0.90) - 1 },
    { zone: 5, low: at(0.90), high: max }
  ];
}

// Smooth before classifying; skip first 45 s of each interval
const smooth = readings => readings.slice(-5).reduce((a, b) => a + b, 0) /
Math.min(5, readings.length);
```

Worked example (checked): age 35, resting HR 60. Tanaka max = 208 − 24.5 = 183.5, rounded to 184. Reserve = 124 bpm. Z2 = 110 to 133 bpm; Z4 starts at 153 bpm.

Resting HR input: take a morning reading from Garmin Connect's resting HR. Optical resting HR agreed closely with ECG in a recent Garmin validation ($r \geq 0.94$; Casanova-Lizón 2026).

For C25K specifically: the goal is finishing the intervals, not hitting a zone. Show zones as information. Alert only when HR stays above the user's Z4 ceiling for 30 seconds.

Sources

All retrieved from PubMed. Abstracts read; full texts not reviewed.

Study	Year	Type	Link
Casanova-Lizón et al., Garmin Elevate v4 resting HR validity, Sensors	2026	Validation, n=23	DOI
Festino et al., Italian Talk Test validation, IJSPP	2026	Validation, n=14	DOI
Inoue et al., Galaxy Watch HR during max treadmill test, JMIR Cardio	2026	Validation, n=55	DOI
Lee et al., Smartwatch HR and energy expenditure validity, Sensors	2026	Comparative, n=62	DOI

Study	Year	Type	Link
Martin et al., Max HR equations across fitness levels, PLoS One	2025	Retrospective, n=230	DOI
Moghaddam et al., Sensor placement and HR accuracy, Sensors	2025	Validation, n=28	DOI
Navalta et al., Tattoos and HR validity, Sensors	2025	Validation, n=25	DOI
Sá Couto et al., Intensity prescription in heart failure, Rev Port Cardiol	2025	Retrospective, n=163	DOI
Kwon et al., Talk test on treadmill vs bike, J Exerc Rehabil	2025	Cross-sectional, n=22	DOI
Milani et al., Predicting HR at ventilatory thresholds, Eur J Prev Cardiol	2024	Multicentre, n=2,868	DOI
Rivera-Köfler et al., Polarised vs other intensity distributions, JSCR	2024	Scoping review	DOI
Weatherwax et al., Personalised vs standardised training, JSSM	2024	RCT, n=109	DOI
Gaskill et al., VT vs HRR and RPE in a large sample, MSSE	2023	Retrospective, n=863	DOI
Kasiak et al., Max HR models in runners and cyclists, J Clin Med	2023	External validation, n=5,069	DOI
Laginestra et al., Cardiac drift and beta-blockade, Am J Physiol	2023	Crossover, n=13	DOI
Mulholland et al., Cardiovascular drift in heat, IJERPH	2023	Experimental, n=8	DOI
Sperlich et al., Elite training intensity distribution, Front Sports Act Living	2023	Review	DOI
Kirton et al., Individualised vs standardised prescription, JSSM	2022	RCT, n=38	DOI
Martini et al., VT2 change after individualised prescription, IJERPH	2022	RCT, n=49	DOI
Vieira et al., Talk test measurement properties, Rev Cardiovasc Med	2022	Systematic review	DOI
Campos et al., Intensity distribution in distance runners, Int J Sports Med	2021	Systematic review	DOI

Study	Year	Type	Link
Shookster et al., Accuracy of age-predicted max HR equations, IJES	2020	Cross-sectional, n=99	DOI